```
<doctype html>

<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width,initial-scale=1">
<title>Trace Route</title>
<style>
*{box-sizing:border-box;margin:0;padding:0}
:root{--bg:#111827;--panel:#1f2937;--panel2:#182230;--border:#374151;--
text:#fff;--muted:#cbd5e1;--blue:#2563eb;--green:#16a34a;--red:#dc2626;
--orange:#d97706;--yellow:#fbbf24}
body{font-family:Arial,sans-serif;background:var(--bg);color:var(--text
);min-height:100vh;transition:.25s}
body.light{--bg:#eef2f7;--panel:#fff;--panel2:#e8eef7;--border:#cbd5e1;
--text:#111827;--muted:#374151}
header{height:58px;padding:0
18px;background:var(--panel);display:flex;justify-content:space-between
;align-items:center;border-bottom:1px solid var(--border)}
main{max-width:850px;margin:auto;padding:20px}
.card{background:var(--panel);border:1px solid
var(--border);border-radius:10px;padding:20px}
button{width:100%;padding:11px;margin:4px
0;border:0;border-radius:6px;color:#fff;background:var(--blue);font-wei
ght:bold;cursor:pointer}
button:hover{opacity:.9}
button:disabled{opacity:.4;cursor:not-allowed}
.grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(170px,1
fr));gap:8px;margin-top:15px}
.green{background:var(--green)}
.red{background:var(--red)}
.orange{background:var(--orange)}
.gray{background:#4b5563}
.info{margin-top:15px;padding:12px;border:1px solid
var(--border);border-radius:7px;line-height:1.6}
.map{height:220px;margin:15px
0;border-radius:8px;background-size:cover;background-position:center}
.battle{display:grid;grid-template-columns:1fr auto
1fr;gap:15px;align-items:center;text-align:center;margin:20px 0}
.unit img{width:145px;height:145px;object-fit:contain}
.vs{font-size:22px;font-weight:bold;color:#fbbf24;min-width:150px;posit
ion:relative}
```

```
.beam{height:8px;width:0;background:#38bdf8;border-radius:10px;box-shadow:0 0 8px #38bdf8,0 0 18px #60a5fa,0 0 30px #fff;animation:shoot .25s ease-out forwards}
@keyframes shoot{
0%{width:0;opacity:0}
20%{width:40px;opacity:1}
100%{width:150px;opacity:0}
}
.hp{width:145px;height:9px;margin:7px auto;background:#000;border-radius:10px;overflow:hidden}
.hp div{height:100%;background:#16a34a}
.msg{margin-top:12px;color:var(--muted);line-height:1.5}
.narrator{display:flex;gap:14px;align-items:center;margin-bottom:18px;padding:13px;background:var(--panel2);border:1px solid var(--border);border-radius:9px}
.narrator img{width:90px;height:90px;object-fit:contain}
.narrator-name{font-weight:bold;color:#fbbf24;margin-bottom:6px}
.narrator-text{line-height:1.5;color:var(--text)}
.loading{position:fixed;inset:0;z-index:20;background:#071525;display:flex;flex-direction:column;align-items:center;justify-content:center;padding:20px;text-align:center}
.loading img{max-width:85%;max-height:55vh;object-fit:contain}
.loading h2{margin:18px 0;letter-spacing:4px}
.loadbox{width:70%;height:16px;border:2px solid #fff;margin-top:15px}
.loadbar{height:100%;width:0;background:#fbbf24}
.loot{margin:15px 0;padding:14px;border:1px solid #16a34a;border-radius:8px;background:#12351f;color:#bbf7d0;font-weight:bold;text-align:center}
.settings{margin-top:15px}
.setting{display:flex;justify-content:space-between;align-items:center;padding:14px;border:1px solid var(--border);border-radius:8px;margin:8px 0;background:var(--panel2)}
.switch{position:relative;width:48px;height:25px}
.switch input{display:none}
.slider{position:absolute;inset:0;background:#6b7280;border-radius:20px;cursor:pointer}
.slider:before{content:"";position:absolute;width:19px;height:19px;left:3px;top:3px;background:#fff;border-radius:50%;transition:.2s}
.switch input:checked+.slider{background:#16a34a}
.switch input:checked+.slider:before{transform:translateX(23px)}
@media(max-width:600px){
.battle{grid-template-columns:1fr}
.vs{transform:rotate(90deg)}
```

```
.beam{transform:rotate(90deg)}
.narrator{align-items:flex-start}
.narrator img{width:75px;height:75px}
}
</style>
</head>

<body>

<header>
<b>TRACE ROUTE</b>
<span id="stats"></span>
</header>

<main>
<div class="card" id="screen"></div>
</main>

<script>
const IMG={
player:"https://uploads.onecompiler.io/44q7nf52a/1786954261039/player_tank.png.png",
drones:[
"https://uploads.onecompiler.io/44q7nf52a/44ygbu2db/drone1.png",
"https://uploads.onecompiler.io/44q7nf52a/44ygbu2db/drone2.png",
"https://uploads.onecompiler.io/44q7nf52a/44ygbu2db/drone3.png",
"https://uploads.onecompiler.io/44q7nf52a/44ygbu2db/drone4.png"
],
assembly:"https://uploads.onecompiler.io/44q7nf52a/1786954226493/assembly_loading.png",
narrator:"https://uploads.onecompiler.io/44q7nf52a/44ygbu2db/narrator1.png",
maps:[
"https://uploads.onecompiler.io/44q7nf52a/1786954240223/map1.png",
"https://uploads.onecompiler.io/44q7nf52a/1786954245412/map2.png",
"https://uploads.onecompiler.io/44q7nf52a/1786954251026/map3.png",
"https://uploads.onecompiler.io/44q7nf52a/1786954256117/map4.png"
]
};

let game={
screen:"home",
hp:100,
```

```
maxHp:100,
atk:12,
def:2,
scrap:2,
wire:1,
battery:0,
parts:0,
bosses:0,
modules:[false,false,false],
enemy:null,
beam:false,
narrator:"Welcome, commander. I will teach you how to play.",
msg:""
};

let settings={
theme:localStorage.getItem("traceTheme")||"dark",
sound:localStorage.getItem("traceSound")!=="off"
};

const screen=document.getElementById("screen");
const stats=document.getElementById("stats");

function applySettings(){
document.body.classList.toggle("light",settings.theme==="light");
localStorage.setItem("traceTheme",settings.theme);
localStorage.setItem("traceSound",settings.sound?"on":"off");
}

function playSound(type){
if(!settings.sound)return;
try{
const AudioContext=window.AudioContext||window.webkitAudioContext;
const ctx=new AudioContext();
const oscillator=ctx.createOscillator();
const gain=ctx.createGain();
let frequency=500;
let duration=.08;
if(type==="attack"){frequency=180;duration=.06}
if(type==="hit"){frequency=100;duration=.06}
if(type==="victory"){frequency=850;duration=.2}
if(type==="battery"){frequency=1100;duration=.12}
oscillator.frequency.value=frequency;
```

```
oscillator.type="square";
gain.gain.value=.035;
oscillator.connect(gain);
gain.connect(ctx.destination);
oscillator.start();
oscillator.stop(ctx.currentTime+duration);
}catch(e){}
}

function say(text){
game.narrator=text;
}

function hpBar(now,max){
let p=max?now/max*100:0;
p=Math.max(0,Math.min(100,p));
return `<div class="hp"><div style="width:${p}%"></div></div>`;
}

function narratorBox(){
return `<div class="narrator"><img src="${IMG.narrator}"
alt="Narrator"><div><div class="narrator-name">NARRATOR</div><div
class="narrator-text">${game.narrator}</div></div></div>`;
}

function updateStats(){
if(["home","win","lose"].includes(game.screen)){
stats.innerHTML="";
return;
}
stats.innerHTML=`HP ${game.hp}/${game.maxHp} | ATK ${game.atk} | DEF
${game.def} | 🔋 Battery ${game.battery}`;
}

function loading(type,done){
let box=document.createElement("div");
box.className="loading";
let
image=type==="assembly"?IMG.assembly:IMG.maps[Math.floor(Math.random()*
IMG.maps.length)];
let text=type==="assembly"?"Preparing the assembly workshop...":"Moving
to a new area...";
```

```
box.innerHTML=`<img src="${image}"
alt="Loading"><h2>LOADING...</h2><p>${text}</p><div
class="loadbox"><div class="loadbar"></div></div>`;
document.body.appendChild(box);

let bar=box.querySelector(".loadbar");
let p=0;

let timer=setInterval(()=>{
p+=5;
bar.style.width=p+"%";

if(p>=100){
clearInterval(timer);

setTimeout(()=>{
box.remove();
done();
},100);
}
},35);
}

function show(name){
game.screen=name;
game.msg="";
game.beam=false;
render();
}

function startGame(){
game={
screen:"map",
hp:100,
maxHp:100,
atk:12,
def:2,
scrap:2,
wire:1,
battery:0,
parts:0,
bosses:0,
modules:[false,false,false],
```

```
enemy:null,
beam:false,
narrator:"Your tank is ready. Explore the area and collect resources.",
msg:""
};

playSound("victory");
render();
}

function home(){
say("Returning home. The map transition is loading.");
loading("map",()=>show("home"));
}

function fight(boss){
let i=Math.floor(Math.random()*IMG.drones.length);
let max=boss?40+game.bosses*20:20;

game.enemy={
name:boss?`Alpha Drone ${game.bosses+1}`:"Scout Drone",
hp:max,
max:max,
atk:boss?10+game.bosses*3:6,
boss:boss,
img:IMG.drones[i]
};

game.beam=false;

say("Enemy detected. Attack the drone.");
playSound("hit");
show("fight");
}

function attack(){
if(!game.enemy)return;

game.beam=true;

game.enemy.hp-=game.atk;

playSound("attack");
```

```
if(game.enemy.hp<=0){
game.enemy.hp=0;
say("Drone destroyed! Collect the loot to receive 1 Battery.");
playSound("victory");
render();

setTimeout(()=>{
game.beam=false;
render();
},300);

return;
}

let damage=Math.max(1,game.enemy.atk-game.def);
game.hp=Math.max(0,game.hp-damage);
game.msg=`You dealt ${game.atk} damage. You took ${damage}.`;

if(game.hp<=0){
say("Your tank has been disabled.");
}else{
say("Keep attacking until the drone is destroyed.");
}

playSound("hit");
render();

setTimeout(()=>{
game.beam=false;
render();
},300);
}

function retreat(){
say("Retreating to a new area.");
loading("map",()=>show("map"));
}

function collectLoot(){
if(!game.enemy)return;

if(game.enemy.boss){
```

```
game.bosses++;
game.parts++;
game.battery++;
say("Alpha Drone defeated. You collected 1 Transmitter Part and 1
Battery.");
}else{
game.scrap+=2;
game.wire++;
game.battery++;
say("Scout Drone defeated. You collected 2 Scrap, 1 Wire, and 1
Battery.");
}

playSound("battery");
loading("map",()=>show("map"));
}

function scavenge(){
game.screen="scan";
say("Search the area for useful resources.");
render();

let p=0;

let timer=setInterval(()=>{
p+=5;

let bar=document.getElementById("scanBar");

if(bar){
bar.style.width=p+"%";
}

if(p>=100){
clearInterval(timer);

if(Math.random()<.5){
game.scrap++;
game.msg="Found 1 Scrap.";
say("You found scrap.");
}else{
game.msg="Nothing useful found.";
say("Nothing useful was found here.");
```

```
}

show("map");
}
},100);
}

function upgrade(type){
if(type==="hp"&&game.scrap>=4&&game.battery>=1){
game.scrap-=4;
game.battery--;
game.maxHp+=20;
game.hp+=20;
say("Armor upgraded. Your tank has more health.");
}else if(type==="atk"&&game.wire>=3&&game.scrap>=2){
game.wire-=3;
game.scrap-=2;
game.atk+=5;
say("Weapons upgraded. Your tank deals more damage.");
}else if(type==="def"&&game.scrap>=5){
game.scrap-=5;
game.def+=2;
say("Defense upgraded. Enemy attacks do less damage.");
}else{
game.msg="Not enough resources.";
say("You do not have enough resources for that upgrade.");
}

playSound("battery");
render();
}

function openAssembly(){
say("Opening the Assembly Workshop.");
loading("assembly",()=>show("assembly"));
}

function build(i){
if(game.parts<1){
game.msg="Need 1 Transmitter Part.";
say("Defeat Alpha Drones to collect transmitter parts.");
render();
return;
```

```
}

game.parts--;
game.modules[i]=true;
playSound("battery");

if(game.modules.every(Boolean)){
say("All modules are complete. Mission accomplished.");
show("win");
}else{
say("Module installed. Build the remaining modules.");
render();
}
}

function setTheme(theme){
settings.theme=theme;
applySettings();
playSound("battery");
render();
}

function setSound(value){
settings.sound=value;
applySettings();

if(value)playSound("battery");
else render();
}

function homePage(){
return `${narratorBox()}<h2>Trace Route</h2><p>Explore the map, collect
resources, upgrade your tank, defeat drones, and assemble the final
system.</p><br><button class="green" onclick="startGame()">Start
Game</button><button class="gray" onclick="show('settings')">⚙
Settings</button>`;
}

function mapPage(){
let map=IMG.maps[Math.floor(Math.random()*IMG.maps.length)];

return `${narratorBox()}<h2>Area</h2><div class="map"
style="background-image:url('${map}')"></div><div class="grid"><button
```

```
onclick="scavenge()">Scavenge</button><button class="orange"
onclick="fight(false)">Scout Drone</button><button class="red"
onclick="fight(true)">Alpha Drone (${game.bosses}/3)</button><button
class="green" onclick="say('Use scrap, wire, and batteries to
strengthen your tank.');show('upgrade')">Upgrade Tank</button><button
onclick="openAssembly()">Assembly Workshop</button><button class="gray"
onclick="show('settings')">⚙ Settings</button><button class="gray"
onclick="home()">Home</button></div><div class="info">Scrap:
${game.scrap}<br>Wire: ${game.wire}<br>🔋 Battery:
${game.battery}<br>Transmitter Parts: ${game.parts}/3</div><div
class="msg">${game.msg}</div>`;
}

function scanPage(){
return `${narratorBox()}<h2>Scanning Area...</h2><p>Searching for
useful resources.</p><div class="loadbox"><div id="scanBar"
class="loadbar"></div></div>`;
}

function fightPage(){
let e=game.enemy;

if(game.hp<=0){
return `${narratorBox()}<h2>Tank Disabled</h2><div class="battle"><div
class="unit"><img src="${IMG.player}" alt="Tank"><p>Player
Tank</p></div><div class="vs">VS</div><div class="unit"><img
src="${e.img}" alt="Drone"><p>${e.name}</p></div></div><button
class="red" onclick="show('lose')">Continue</button>`;
}

if(e.hp<=0){
let loot=e.boss?"1 Transmitter Part + 1 Battery":"2 Scrap + 1 Wire + 1
Battery";

return `${narratorBox()}<h2>Victory</h2><div class="battle"><div
class="unit"><img src="${IMG.player}"
alt="Tank">${hpBar(game.hp,game.maxHp)}<p>Player Tank</p></div><div
class="vs">${game.beam?'<div class="beam"></div>':'WIN'}</div><div
class="unit"><img src="${e.img}"
alt="Drone"><p>${e.name}</p></div></div><div class="loot">🔋 Loot:
${loot}</div><button class="green" onclick="collectLoot()">Collect
Loot</button>`;
}
```

```
return `${narratorBox()}<h2>${e.name}</h2><div class="battle"><div
class="unit"><img src="${IMG.player}"
alt="Tank">${hpBar(game.hp,game.maxHp)}<p>Player Tank</p></div><div
class="vs">${game.beam?'<div class="beam"></div>':'VS'}</div><div
class="unit"><img src="${e.img}"
alt="Drone">${hpBar(e.hp,e.max)}<p>${e.name}</p></div></div><div
class="grid"><button class="red" onclick="attack()">⚡
Attack</button><button class="gray"
onclick="retreat()">Retreat</button></div><div
class="msg">${game.msg}</div>`;
}

function upgradePage(){
return `${narratorBox()}<h2>Upgrade Tank</h2><p>Spend resources to
improve your tank.</p><div class="info">Scrap: ${game.scrap}<br>Wire:
${game.wire}<br>🔋 Battery: ${game.battery}</div><div
class="grid"><button onclick="upgrade('hp')">+20 HP<br>4 Scrap + 1
Battery</button><button onclick="upgrade('atk')">+5 ATK<br>3 Wire + 2
Scrap</button><button onclick="upgrade('def')">+2 DEF<br>5
Scrap</button></div><br><button class="gray"
onclick="show('map')">Back</button><div class="msg">${game.msg}</div>`;
}

function assemblyPage(){
return `${narratorBox()}<h2>Assembly Workshop</h2><p>Build all three
modules to complete the mission.</p><div class="info">Transmitter
Parts: ${game.parts}/3<br>🔋 Batteries: ${game.battery}</div><div
class="grid"><button onclick="build(0)"
${game.modules[0]?"disabled":""}>${game.modules[0]?"Module A ✓":"Build
Module A"}</button><button onclick="build(1)"
${game.modules[1]?"disabled":""}>${game.modules[1]?"Module B ✓":"Build
Module B"}</button><button onclick="build(2)"
${game.modules[2]?"disabled":""}>${game.modules[2]?"Module C ✓":"Build
Module C"}</button></div><br><button class="gray"
onclick="loading('map',()=>show('map'))">Back</button>`;
}

function settingsPage(){
return `${narratorBox()}<h2>Settings</h2><div class="settings"><div
class="setting"><span>☀ Light Mode</span><label class="switch"><input
type="checkbox" ${settings.theme==="light"?"checked":""}
onchange="setTheme(this.checked?'light':'dark')"><span
```

```
class="slider"></span></label></div><div class="setting"><span>🌙 Dark
Mode</span><label class="switch"><input type="checkbox"
${settings.theme==="dark"?"checked":""}
onchange="setTheme(this.checked?'dark':'light')"><span
class="slider"></span></label></div><div class="setting"><span>🔊 Sound
Effects</span><label class="switch"><input type="checkbox"
${settings.sound?"checked":""} onchange="setSound(this.checked)"><span
class="slider"></span></label></div></div><br><button class="gray"
onclick="show('${game.screen==='settings'&&game.hp>0?'map':'home'}')">B
ack</button>`;
}

function render(){
updateStats();

let html="";

switch(game.screen){
case"home":
html=homePage();
break;

case"map":
html=mapPage();
break;

case"scan":
html=scanPage();
break;

case"fight":
html=fightPage();
break;

case"upgrade":
html=upgradePage();
break;

case"assembly":
html=assemblyPage();
break;

case"settings":
```

```
html=settingsPage();
break;

case"win":
html=`${narratorBox()}<h2>Mission Complete</h2><p>All three modules
have been assembled.</p><br><button class="green"
onclick="home()">Return Home</button>`;
break;

case"lose":
html=`${narratorBox()}<h2>Mission Failed</h2><p>Your tank was
disabled.</p><br><button class="red" onclick="startGame()">Try
Again</button><button class="gray" onclick="home()">Home</button>`;
break;
}

screen.innerHTML=html;
}

applySettings();
render();

</script>

</body>
</html>
```

Link to game: https://trace-route-3.oneapp.dev/